

Fueling Expectations: The Causal Impact of Gas Prices on Inflation Expectations

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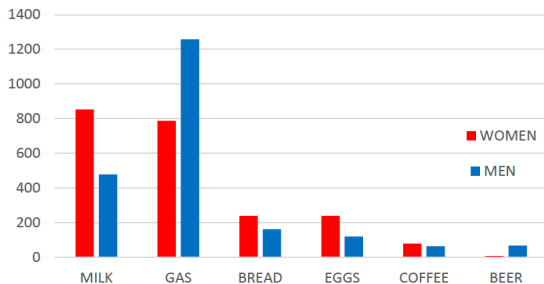
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Households Inflation Expectations and Gas Prices

- Inflation expectations play a pivotal role in monetary policy. Real interest rate, $r_t = i_t - \mathbb{E}_t \pi_{t+1}$, matters in determining households' consumption or savings

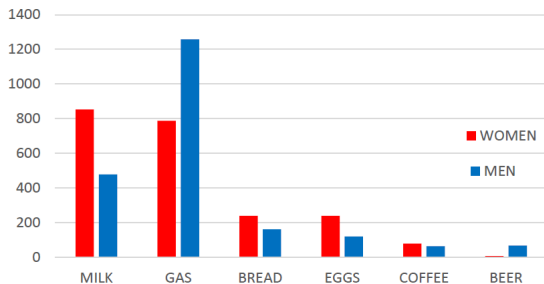
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 - ▶ Coibion and Gorodnichenko (2015): Gas prices are salient
 - ▶ D'Acunto and Weber (2022): Which goods' prices came to mind when forming expectations?



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- Most studies rely on time-series variation or structural identification. Endogeneity bias concerns.

Quasi-experimental Evidence: Gas Tax Holidays



- Supply disruptions caused by the Covid pandemic and the Russia-Ukraine war led to soaring gas prices
- While 21 states formally considered temporary gas tax suspensions, only five states implemented them.

This Paper: Causal Impact of Gas Prices on Inflation Expectations

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- Use a difference in differences strategy to measure the impact of gas prices on inflation expectations and consumption

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- No stimulative impact on consumption

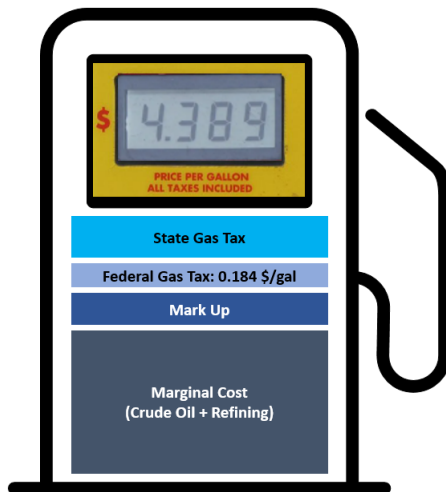
Previous literature and Contribution

- Gas prices and household inflation expectations in the US
 - ▶ Coibion and Gorodnichenko (2015); Binder (2018); Kilian and Zhou (2022)
 - ▶ Bring new quasi-experimental evidence to study the effect of gas prices on beliefs
- Validity of FIRE models
 - ▶ Study unique short-lived, temporary shock
 - ▶ Backward-looking belief formation, deviation from FIRE
- Effects of consumption tax on inflation expectations
 - ▶ Correia, Farhi, Nicolini, and Teles (2013)
 - ▶ D'Acunto, Hoang, Weber (2022); Bachmann, Born, Goldfayn-Frank, Kocharkov, Luetticke, and Weber (2022)
 - ▶ Demonstrates possible alternative policy levers to impact household beliefs

Gas Tax Holiday and Identification Strategy

State-level Gas Tax

State-level gas taxes ranges from as low as 14 cents per gallon to as high as 62 cents per gallon.



Gas Tax Holiday in the United States

- 21 U.S. states formally proposed temporary gas tax suspensions, only five states implemented them.
- No implementation lags for Maryland, Georgia, and Connecticut

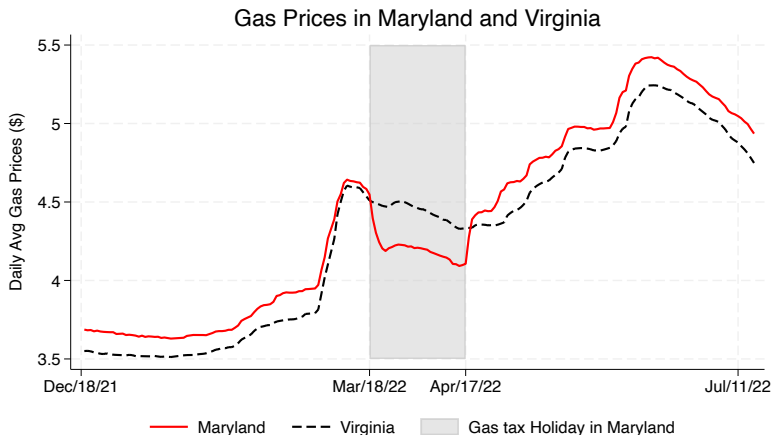
States	Govenor	Gas Tax Cut	Period	Signed
Maryland	Larry Hogan (R)	36.1¢/gal	Mar 18 - Apr 17, 2022	Mar 18
Georgia	Brian Kemp (R)	29.1¢/gal	Mar 18, 2022 - Jan 10, 2023	Mar 18
Connecticut	Ned Lamont (D)	25¢/gal	Apr 1 - Dec 31, 2022	Mar 24
New York	Kathy Hochul (D)	16¢/gal	Jun 1 - Dec 31, 2022	Apr 7
Florida	Ron DeSantis (R)	25.3¢/gal	Oct 1 - 31, 2022	Jul 14

Identification Strategy

- Contemporaneous bias: Russia-Ukraine war disrupted supply chains, elevating gas prices and inflation expectations
- Difference-in-Differences Specification: Compare states that introduced a gas tax holiday to those that did not at the same point in time
- Include time fixed effects, partialling out the impact of changes in the national or global macroeconomic environment
- Identifying assumption:
 - ▶ Inflation expectations in tax holiday states would have followed parallel trends to control states in absence of the policy

Gas Tax Cut in Maryland vs. Virginia

- In March, Virginia's Gov. Youngkin and Maryland's Gov. Hogan pushed for tax suspensions.
- Virginia's proposal was rejected by Democrats in the state Senate
- Maryland's bipartisan effort led to a gas tax suspension



Other states considered tax holiday but did not implement

For example, in Michigan

- House Bill 5570, introduced by state Rep. Steve Johnson (R), would put a halt on the state's 27.2 cents-per-gallon tax from April 1 until Sept. 30.
- The Michigan House voted on 3/9/2022 to pass a bill
- Passed Senate on 3/15/2022.
- Gov. Gretchen Whitmer (D) vetoed this legislation on 4/1/2022.

State-level Conditions in Gas Tax Holiday states vs. others

Macroeconomic conditions across three groups of the states prior to gas tax holiday are similar.

	(1)	(2)	(3)
	States		
	Implemented	Did Not Implement	
Average from 2021M2 to 2022M2		Considered	Did Not Consider
Unemployment Rate (%)	5.26 (0.18)	5.15 (0.09)	4.21 (0.07)
Growth Rate of Real GDP (%)	6.26 (0.37)	5.74 (0.25)	5.00 (0.18)
Growth Rate of Nominal GDP (%)	10.56 (0.39)	11.57 (0.23)	12.27 (0.20)
Inflation rate (%)	4.73 (0.21)	4.90 (0.16)	4.64 (0.08)
1 year ahead Inflation Expectations (%)	5.15 (0.14)	5.03 (0.10)	5.25 (0.08)
Growth Rate of Consumption (%)	16.80 (1.32)	17.08 (0.68)	16.98 (0.47)
The Number of States	5	16	29

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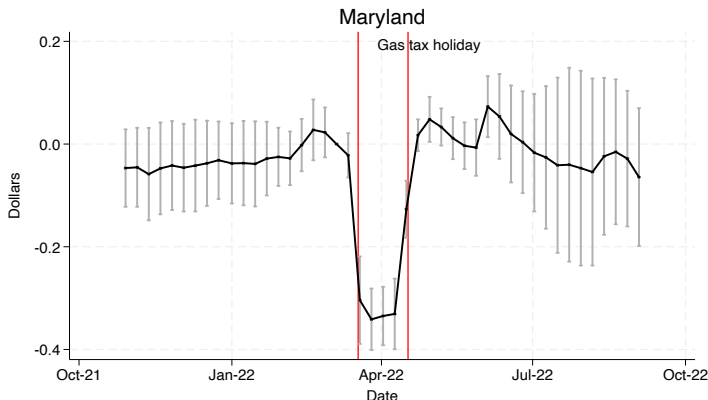
Pass-through to Retail Gas Prices

Pass-through to Retail Gas Prices in Maryland

- Use daily state-level gas prices data from AAA
- Estimate the pass-through to retail gas prices using

$$P_{st} = \alpha_s + \beta_t + \lambda_{st} \times D_s + \epsilon_{st}$$

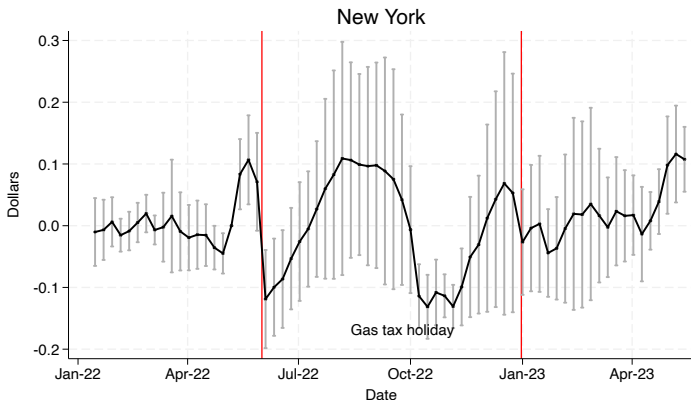
- Gas prices in Maryland are 30 cents lower than neighboring states



Control states: PA, DE, VA, and WV.

Pass-through to Retail Gas Prices in New York

- Negligible impacts on gas prices in New York relative to neighboring states



Control states: PA, DE, NH, and NJ.

Pass-through to Retail Gas Prices

- Significant variation in pass-through rates across states.

$$P_{st} = \alpha_s + \beta_t + \gamma_s D_{st} + \epsilon_{st}$$

	(1)	(2)	(3)	(4)	(5)
	Daily Gas Prices (dollars)				
	MD	GA	CT	NY	FL
Gas Tax Holiday (D_{st})	-0.297*** (0.0323)	-0.178*** (0.0158)	-0.155*** (0.0309)	-0.00341 (0.0390)	-0.204*** (0.00697)
State FE	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes
Observations	4,315	3,452	4,315	4,315	6,041
R^2	0.99	1.00	0.99	0.99	1.00
Gas tax cut (dollar)	0.36	0.29	0.25	0.16	0.25
Passthrough	0.82	0.61	0.62	0.02	0.80
Passthrough 95% CI	[0.65,1.00]	[0.51,0.72]	[0.38,0.86]	[-0.46,0.50]	[0.75,0.86]
Percent change (%)	-7.58	-4.79	-3.74	-0.07	-4.82
95% CI	[-9.19,-5.97]	[-5.63,-3.96]	[-5.21,-2.28]	[-1.68,1.54]	[-5.14,-4.49]

Note: Standard errors clustered by state and week are indicated in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Household Inflation Expectations

State-level Household Inflation Expectations data

- Survey of Consumer Expectations (SCE) from NY-Fed
 - ▶ SCE surveys about 1,300 respondents per month
 - ▶ An average of about 25 respondents per state each month
- In order to improve statistical power, we conducted an additional Online survey
 - ▶ Treated states (Maryland, Georgia, New York, and Florida) as well in neighboring states that did not cut the gas tax around the holiday
 - ▶ Our survey instrument mirrors that of the questions asked in the SCE
 - ▶ Panel data
 - ▶ Construct sampling weights to match state-level demographic features

State-level specification: DID

Compared to neighboring control states

- Respondents in a treated state reduced inflation expectations
- Deviation from FIRE model

$$E_t^i \pi_{s,t+1} = \alpha_i + \beta_t + \gamma X_{it} + \beta D_{st} + \epsilon_{st}$$

	(1)	(2)	(3)	(4)	(5)
		Expected inflation rate (density)			
Gas Tax Holiday (dummy)	-1.365** (0.448)	-1.449** (0.316)	-2.065** (0.472)	-0.0684 (0.233)	-0.0171 (0.0789)
State	MD	GA	CT	NY	FL
Sampling Weight	Yes	Yes	Yes	Yes	Yes
Individual Fixed Effect	Yes	Yes	Yes	Yes	Yes
Time Fixed Effect	Yes	Yes	Yes	Yes	Yes
Survey Tenure Fixed Effect	Yes	Yes	Yes	Yes	Yes
Data	Combined	Combined	Combined	Combined	Combined
Observations	1,881	2,251	2,312	3,494	6,241
R ²	0.71	0.64	0.66	0.71	0.71

Note: Standard errors clustered by state and week are indicated in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

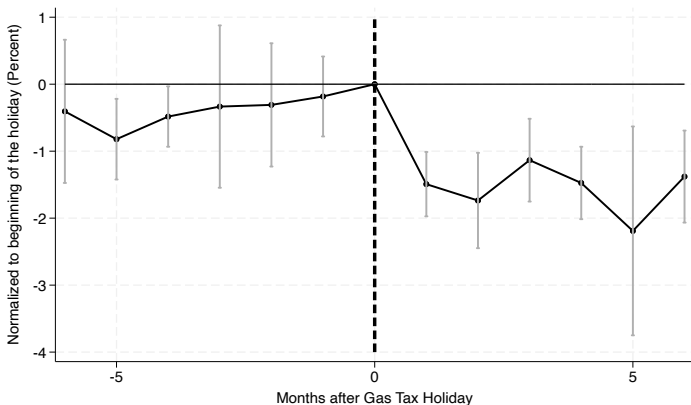
Attention

Pooled Specification: Event Study Plot

- Stacked DID

$$E_t^i \pi_{e,t+1} = \alpha_{ei} + \beta_{et} + \gamma X_{eit} + \beta D_{eit} + \epsilon_{eit}$$

- No evident pre-trend. After gas tax holiday in MD, GA, and CT, a noticeable decline in household inflation expectations



Pooled Specification: Inflation expectations

- Gas tax holiday reduced inflation expectations by 0.31 percentage points on average.
- States experiencing larger declines in gas prices show more substantial decreases in inflation expectations.

	(1)	(2)	(3)
	One-year-ahead expected inflation rate (density)		
Gas Tax Holiday (dummy)	-0.307 (0.252)	-1.351*** (0.221)	0.235 (0.232)
Gas Tax Holiday × Percent Gas Price Change			0.132* (0.0685)
Sample	All	Excl. FL, NY	All
Sampling Weight	Yes	Yes	Yes
Experiment × Individual FE	Yes	Yes	Yes
Experiment × Time FE	Yes	Yes	Yes
Experiment × Survey Tenure FE	Yes	Yes	Yes
State-level controls	Yes	Yes	Yes
Data	Combined	Combined	Combined
Observations	68,463	41,122	68,463
R^2	0.67	0.66	0.67

Note: Standard errors clustered by state and week are indicated in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Comparisons with previous literature

- Impact of a 1 pp decrease in gas prices on inflation expectations:
 - ▶ Our study: 0.13 pp
 - ▶ Coibion and Gorodnichenko (2015): 0.016 pp
 - ▶ Kilian and Zhou (2022): 0.03 pp

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- Replicated analysis of CG for our sample period: 2021M12-2023M1

	$\pi_{it}^e - \pi_{it-1}^e$
$\log\left(\frac{\text{Gas}_t}{\text{Gas}_{t-1}}\right) * 100$	0.0364*** (0.00366)
Observations	16,072
R^2	0.01

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Observations	16,072
R^2	0.01

- Highlights the identification advantages
 - ▶ Time fixed effects
 - ▶ Quasi-experiment: exogenous gas price variation

Robustness Results

- Robust to control states: a tax holiday was formally proposed but not implemented
- Placebo test: Placebo = 1 for those states proposed policy in the non-implementing states

	(1)	(2)
	Expected inflation rate	
Gas Tax Holiday (Dummy)	-0.461*	(0.274)
Sampling Weight	Yes	Yes
Experiment $\times$ Individual FE	Yes	Yes
Experiment $\times$ Time FE	Yes	Yes
Experiment $\times$ Tenure FE	Yes	Yes
State-level controls	Yes	Yes
Observations	37,482	55,383
R^2	0.65	0.68

Note: Standard errors clustered by state and week are indicated in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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	(1)	(2)
	Expected inflation rate	
Gas Tax Holiday (Dummy)	-0.461*	
	(0.274)	
Placebo (Dummy)		-0.000828
		(0.410)
Sampling Weight	Yes	Yes
Experiment $\times$ Individual FE	Yes	Yes
Experiment $\times$ Time FE	Yes	Yes
Experiment $\times$ Tenure FE	Yes	Yes
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Public Policy Communication

Information treatment

- Pass-through to retail gas prices is market outcome, not directly managed by policymakers.
- Alternative policy tool: public policy communication.
- Randomized Experiments
 - ▶ Information about the Gas Tax Holiday provided randomly to respondents in treated states in the first wave of survey
 - ▶ The size and duration of Gas Tax Cut
- After information treatment, solicit inflation expectations and consumption sentiments again
- Study the revision of inflation expectations and consumption sentiments

Effects of Information Treatment

- Providing information reduces inflation expectations
- A larger effect for respondents surveyed before the tax holiday commenced.

$$E_t^{i,post} \pi_{s,t+1} - E_t^{i,pre} \pi_{s,t+1} = \alpha_s + T_{ist} + \text{control}_{ist} + \epsilon_{ist}$$

	(1)	(2)	(3)
	Revisions in inflation expectations		
Information treatment	-0.704* (0.371)	0.174 (0.980)	-0.890** (0.426)
		Survey Implementation	
		During	Before
Sample fixed effect	Yes	MD1,FL2	NY1,FL1
Control for demographics	Yes	Yes	Yes
Robust	Huber	Huber	Huber
Observations	1,220	209	1,011
R ²	0.05	0.15	0.08

Note: * p<0.10, ** p<0.05, *** p<0.01.

Consumption Sentiments

- Respondents with information treatment are less inclined to buy durable goods

$$\text{Good Time to Buy}_{ist}^{e,post} - \text{Good Time to Buy}_{ist}^{e,pre} = \alpha + T_{ist} + \text{control}_{ist} + \epsilon_{ist}$$

	(1)	(2)	(3)
	Revisions in Good Time to Purchase		
	House	Car	Durables
Information treatment	-0.152 (0.105)	-0.306*** (0.117)	-0.104 (0.0960)
Sample Fixed Effect	Yes	Yes	Yes
Control for demographics	Yes	Yes	Yes
Observations	471	473	438
R^2	0.05	0.08	0.05

Consumption

Consumption during Gas Tax Holiday

- Debit/credit card weekly consumption data from Opportunity Insights
 - ▶ 10% of debit and credit card spending in the U.S.
 - ▶ Debit and credit card transactions account for 40% of total consumption.

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 - ▶ 10% of debit and credit card spending in the U.S.
 - ▶ Debit and credit card transactions account for 40% of total consumption.
- Changes in gas spending:
 - ▶ Inelastic gas demand
 - ▶ A reduction in gas prices results in decreased total expenditure on gas
- Changes in non-gas spending:
 - ▶ Discretionary income effect: Boost non-gas spending (+)
 - ▶ Changes in inflation expectation:
 - ★ Intertemporal substitution effect (-)
 - ★ Sentiment effect (+)

State-level specification: DID

No stimulative impact on consumption

$$\ln(C_{st}) = \alpha_s + \beta_t + \gamma D_{st} + \epsilon_{st}$$

	(1)	(2)	(3)	(4)	(5)
	Log Spending $\times$ 100				
Gas Tax Holiday (Dummy)	-1.306** (0.379)	0.130 (0.238)	-1.155 (0.945)	1.964*** (0.268)	-0.508 (0.407)
State	MD	GA	CT	NY	FL
State FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
Observations	160	288	355	290	231
R^2	0.80	0.75	0.50	0.65	0.85

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Categories

Event

Pooled Specification: Categories of Consumption

Except New York and Florida

No stimulative impact on consumption

$$\ln(C_{est}) = \alpha_{es} + \beta_{et} + \gamma D_{est} + \epsilon_{est},$$

	(1)	(2)	(3)	(4)
		Log Spending $\times$ 100		
		Goods		Services
	All	Durables	Non-durables	
Gas Tax Holiday (Dummy)	-0.645*** (0.171)	0.735 (0.459)	-1.244** (0.565)	-1.799* (0.935)
Experiment $\times$ State FE	Yes	Yes	Yes	Yes
Experiment $\times$ Time FE	Yes	Yes	Yes	Yes
State-level Controls	Yes	Yes	Yes	Yes
Observations	8,050	8,050	8,050	8,050
R^2	0.89	0.76	0.76	0.86

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Main

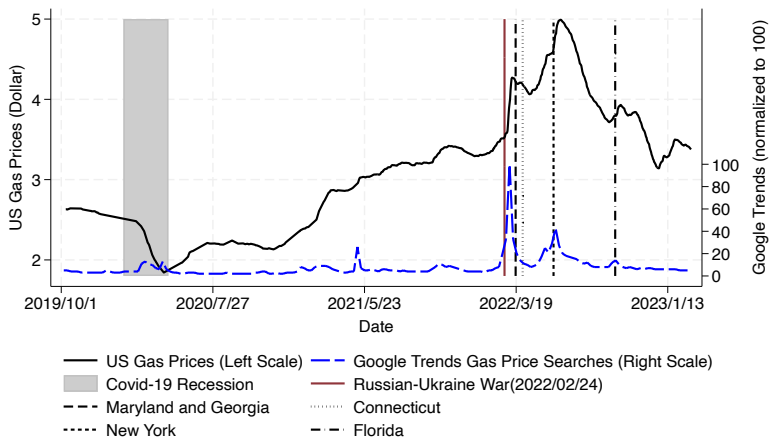
Conclusion

Conclusion

- Provide novel quasi-experimental variation in gas prices
- To study causal impact of gas prices on inflation expectations
- Temporary decreases in gas taxes lead to reductions in household inflation expectations
- The effects are most pronounced in states where the tax reduction resulted in substantial changes in gas prices.
- No stimulative effect on consumption

Gas Prices and Google Trends

Google Searches peak prior to the implementation of the policies in Maryland, Georgia, and Connecticut



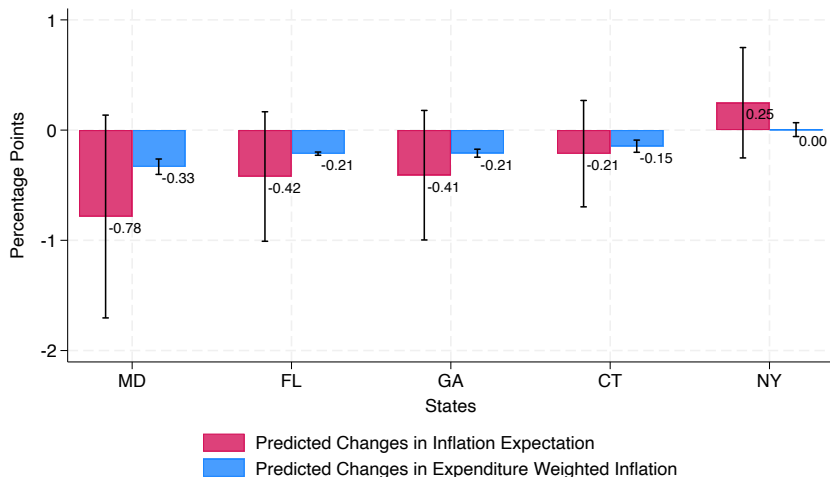
Effect larger than would be expected from gas expenditure share

Consider Maryland

- With an average pass-through rate of 0.83, there was a corresponding 0.78 percentage point reduction in inflation expectations.
- Maryland experienced 29.9 cents cut in gas prices
- The 3 month average gas prices was before gas tax holiday \$3.5 per gallon, which is; 7.7% reduction in gas prices
- Gas expenditure is about 4.3% of total expenditure in 2022
- The 3 month average 1-year ahead inflation expectation was about 8% before gas tax holiday
- Expenditure weighted in gas prices would anticipate only a 0.033 pp fall

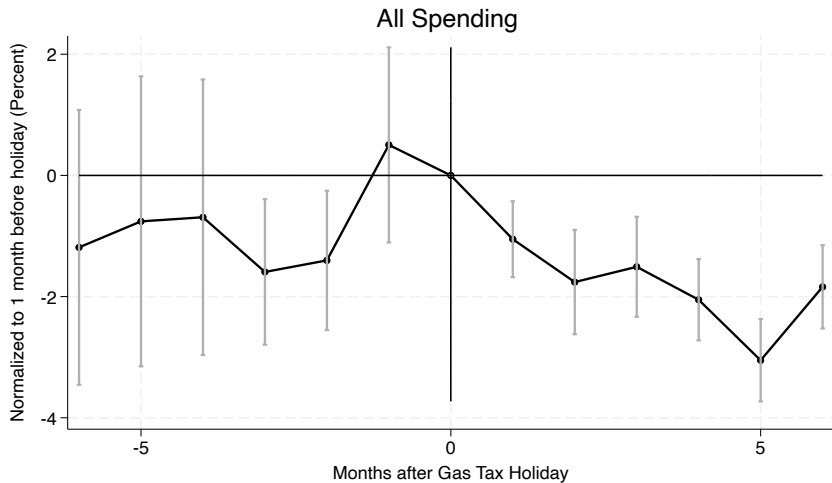
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Pooled Specification: Event Study Plot

Except New York and Florida



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